

AEMB: Canonical Protocol-Graph End-to-End Conformance for Hybrid Financial Protocols

Twelve Parent FIA Cases, Additive Venue and Legal Controls, and Registered Evidence Derivations

Maksym Nechepurenko

ForesightFlow, Devnull FZCO, Dubai, United Arab Emirates

maksym@devnull.ae | [ORCID: 0000-0002-9515-8841](https://orcid.org/0000-0002-9515-8841)

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Abstract

AEMB v0.1.0 provides 38 deterministic scenarios for event-margin accounting, venue-adapter boundaries, and operational faults. This standalone paper specifies AEMB v0.2, an additive conformance extension built around twelve parent Financial Interaction Assertions (FIA-001–FIA-012), five prospective venue/legal cases outside that denominator, and proposed negative-control families. The abstract case body is manifest-independent; deployment manifest and evidence-policy version are bound at the case-manifest layer, while source provenance is registered separately. Seven retained evidence layers carry explicit derivation topology rather than a blanket claim of independence. Promotion predicates are typed as positive, terminal-closure, absence-negative-control, recomputed-comparison, or proposed multi-root-corroboration gates. FIA-011 contains two mandatory subcases: stale capability before admission and book-epoch change after position opening while debt remains positive. The reference Phase 17B corpus contains one all-12 cohort with 84 retained `ACTUAL_EVIDENCE` positions, two clean materializations sharing one byte-identical file index, a deterministic strict archive accepted by a separate verifier, and a selected P01–P17 replay whose P17 output is content-bound to that archive. AEMB v0.2 therefore extends deterministic terminal-state testing into cohort, derivation, packaging, and replay conformance. The article is evidence-integrated; the additive public dataset release remains a separate versioned artifact requiring schema/package freeze and independent package review.

Keywords: benchmark; smart contracts; end-to-end conformance; provenance; event markets; reproducibility; mutation testing; deterministic datasets; evidence packages.

1 Introduction

A benchmark can test a function, a contract, an adapter, or an operational control without establishing that the full application exercised the intended financial path. AEMB v0.1.0 deliberately focuses on deterministic conformance: eight vertical-slice cases, fifteen venue-adapter boundary cases, and fifteen operational/fault rehearsals [Nechepurenko, 2026a]. Its architecture-neutral objects are suitable for regression, cross-language canonicalization, and negative controls, but they do not encode a complete hybrid execution proof.

Phase 17B introduces a stronger assurance target. A user command should be traceable through an authenticated API, durable orchestration, an emulated venue, manifest-authorized canonical contracts, ordered chain events, an append-only journal, application projections, a browser surface, and an audit reconstruction. The test must also reject known-bad linkages: a mutated selector, copied fixture hash, stale projection, duplicated evidence, or second financial transaction after recovery. A benchmark for this target must represent not only expected state but the evidence required to justify a claim.

AEMB v0.2 is designed as an additive extension, not a rewrite. The v0.1.0 semantic identities remain unchanged; a new FIA collection introduces twelve canonical protocol-graph end-to-end cases. Version 0.2 also adds a manifest-bound evidence model and a conjunctive verdict that makes missing layers visible.

Contributions.

- (C1) A twelve-parent FIA collection spanning the critical local financial workflows of a modular event-margin protocol, with mandatory FIA-011A/B subcases and no denominator drift.
- (C2) A manifest-independent abstract case body, a separate case-manifest binding, and a separate source-provenance registration object.
- (C3) Seven retained evidence-layer requirements with registered derivation roots and claim-specific proposed multi-root corroboration; that gate was not evaluated for the Phase 17B reference cohort.
- (C4) Five additive venue/legal cases outside the FIA denominator: timer restart, book reset, post-close lock, pause modes, and `LEGAL_NO_GO`.
- (C5) Per-case verdicts that preserve partial and blocked outcomes rather than collapsing them into a score.
- (C6) Manifest, case, collection, evidence-package, and release identity definitions.
- (C7) Mandatory mutation families for selectors, receipts, event order, projections, packages, replay, and governance.
- (C8) A reproducible package protocol with multi-language validation, two clean runs, historical replay, SBOM, privacy controls, and independent review.
- (C9) A versioning and governance policy for an additive AEMB release and a future relationship to AVET traces.

2 AEMB v0.1.0 baseline

The current public AEMB release contains 38 deterministic scenarios:

$$8 \text{ VS} + 15 \text{ VA} + 15 \text{ OR} = 38.$$

The collections have different purposes:

- VS tests financial vertical-slice transitions and terminal-state identities;
- VA tests venue-adaptor capabilities, ordering, evidence, and lifecycle boundaries;
- OR tests operational faults, recovery, observability, and policy boundaries.

AEMB v0.1.0 is synthetic, deterministic, and explicitly not a production-safety certificate, live-volume measure, user-behaviour dataset, yield forecast, or regulatory approval. It also does not populate claim-level traceability identifiers. The new FIA collection preserves those boundaries while adding evidence-package conformance.

The v0.2 top-level target is therefore

$$38 \text{ existing cases} + 12 \text{ parent FIA cases} + 4 \text{ additive venue cases} + 1 \text{ legal case} = 55.$$

FIA-011A and FIA-011B are mandatory subcases inside FIA-011 and do not increase the parent denominator. Registered negative controls likewise do not increase the top-level case count.

The identity hierarchy and non-automatic verdict progression are illustrated in figs. 1 and 2.

3 Design goals

The v0.2 extension follows eight goals.

- (D1) **Additivity.** Existing v0.1.0 cases and digests remain unchanged.
- (D2) **Manifest binding.** A case cannot pass against an unregistered contract graph.
- (D3) **Case-wise transparency.** Every safety-critical failure remains visible.
- (D4) **Registered derivation and explicit root ancestry.** Expected, observed, and reconstructed layers retain separately identified inputs, registered derivations, and declared root ancestry; independent runtime participation is claimed only under an applicable separately evaluated multi-root corroboration gate; the Phase 17B status of that stronger property is `NOT_PROVEN`.
- (D5) **Negative-control completeness.** Known-bad mutations are first-class benchmark items.
- (D6) **Reproducible packaging.** The release can be rebuilt and independently inspected.
- (D7) **Rights and privacy safety.** No secret, personal, production, or raw third-party venue data is required.
- (D8) **Architecture tolerance.** An implementation need not copy Axient’s internal code, but it must satisfy the declared authority and evidence predicates for the selected conformance profile.

4 FIA collection

The canonical dataset identifiers are proposed as `FIA-001` through `FIA-012`; human-facing prose may retain `FIA-01` through `FIA-12`. Table 1 defines the collection.

Table 1: AEMB v0.2 FIA cases.

Case	Scenario	Conformance focus
FIA-001	successful admission	risk approval, atomic position/debt origination, evidence and projection identity
FIA-002	exact 2× admission	discrete leverage policy, exact calldata, same canonical graph
FIA-003	risk rejection	typed cause, zero financial mutation, audit visibility
FIA-004	partial then final settlement	admitted partial and final evidence, bounded debt application, lifecycle closure
FIA-005	residual and collateral release	one-time release after canonical debt-clearance eligibility
FIA-006	liquidation to hard-flat	authorized liquidation request/execution and hard-flat transition
FIA-007	liquidation/waterfall	liquidation record, reserve-LBP-Senior ordering, conservation
FIA-008	evidence conflict	canonical exception/quarantine and zero debt mutation
FIA-009	Senior withdrawal queue	<code>CreditPool</code> queue, loss participation, FIFO/service/burn ordering
FIA-010	exactly-once recovery	process restart, one financial transaction, projection rebuild
FIA-011	conjunctive capability transition	FIA-011A stale-before-admission and FIA-011B book-epoch change with $D_t > 0$; both mandatory

Case	Scenario	Conformance focus
FIA-012	pause/recovery	frozen target set, guardian pause, pause-policy mode, timelock queue/execute/unpause

The scenarios are intentionally heterogeneous. A successful admission case requires a committed financial transition, whereas a risk rejection case requires the absence of that transition plus a typed, auditable reason. Both can qualify as end-to-end if their complete evidence chains are present.

5 Conformance profiles

AEMB v0.2 supports three profiles so that architecture differences are explicit rather than silently ignored.

Definition 5.1 (Reference-graph profile). *The reference-graph profile evaluates the Axient canonical module partition and requires the declared storage authorities, controller boundaries, settlement evidence registry, queues, loss waterfall, pause targets, and seven evidence layers.*

Definition 5.2 (Equivalent-authority profile). *An implementation may use different module names or deployment topology if it supplies a machine-readable authority map proving that each financial domain has one authoritative owner, no test or orchestration layer duplicates financial state, and every FIA invariant is preserved.*

Definition 5.3 (Evidence-only profile). *The evidence-only profile evaluates the manifest, correlation, derivation, package, replay, and review machinery over an implementation-defined financial workflow. It does not claim Axient financial-protocol conformance.*

A public result must state the profile. Passing the evidence-only profile cannot be reported as passing the reference-graph financial profile.

6 Detailed FIA specifications

This section provides the normative intent for each case. The Phase 17B reference identities are fixed by the public evidence companion; the additive AEMB v0.2 dataset payloads and release manifest remain subject to their own versioned dataset freeze.

6.1 FIA-001: successful admission

Preconditions. A manifest-authorized market and pool are active; the role is Trader; collateral and risk limits admit the request; the venue capability is fresh; leverage is within the $2\times$ local scope.

Action. Submit one typed position-open command. The canonical path consumes a risk approval, opens the position, creates debt, and binds the resulting position to the command correlation.

Required invariants.

- exactly one approval is consumed;
- one position and one debt record are created;
- collateral and debt terms match the command;
- no controller-owned financial state is created;
- all seven evidence layers identify the same command and manifest.

Negative controls. Wrong role, stale manifest, wrong selector, reused approval, and copied fixture hash.

6.2 FIA-002: exact 2× admission

This case specializes FIA-001 to the exact discrete leverage policy. The command and calldata must prove that gross exposure equals twice trader collateral and that no hidden 3× or 5× executable tier is enabled. The benchmark stores exact integer/base-unit values to avoid floating-point ambiguity.

6.3 FIA-003: risk rejection

Preconditions. One registered risk predicate fails, for example pool limit, stale capability, collateral rule, or market eligibility.

Expected result. The command is rejected with a typed reason. No approval is consumed, no position/debt is created, no collateral moves, and every financial before/after projection is equal. The absence of a financial receipt can be correct if the rejection occurs before chain submission, but the API, journal, browser, and audit layers must still bind the typed no-mutation outcome.

6.4 FIA-004: partial then final settlement

The case retains a lifecycle containing a bounded partial settlement followed by distinct final admitted evidence. At the partial stage, the admitted amount reduces debt without enabling residual or collateral release; replay of the same receipt creates zero additional financial delta. At the final stage, the registered final evidence establishes the settlement and debt-clearance condition used by FIA-005. The case body therefore permits an intermediate non-final state but does not require the overall scenario to remain non-final.

6.5 FIA-005: residual and collateral release

Starting from the debt-clearance eligibility established by FIA-004, trader residual and collateral are released exactly once and only after the canonical getters and lifecycle state permit release. The case does not admit new settlement evidence or perform another debt reduction; it verifies the ownership transition after settlement.

6.6 FIA-006: liquidation to hard-flat

An authorized liquidation request and execution are bound to the position/evidence state, followed by the canonical hard-flat transition. The case retains the request and execution identity and remains distinct from the reserve-LBP-Senior allocation performed in FIA-007.

6.7 FIA-007: liquidation and waterfall

The scenario begins with a recognized shortfall and configured capacities. It must show:

$$\ell = a_0 + a_R + a_J + a_G + a_S + \ell_{res},$$

with the registered order. The canonical reserve, LBP, and Senior modules provide before/after state; the liquidation manager records request, execution, and allocation digest; the storage-free controller retains no balance. Wrong priority, double-counting, or non-conservation is a failure.

6.8 FIA-008: evidence conflict

Two content-incompatible observations share a replay/conflict domain. The registry opens the canonical exception; debt, residual, collateral, and pool state remain unchanged. The audit record must distinguish conflict from benign replay.

6.9 FIA-009: loss-participating withdrawal queue

A Senior provider queues **CreditPool** shares, then a registered loss occurs before service. The queued shares remain in supply and absorb their pro-rata impairment. Service and burn occur later under FIFO and available-cash policy. A fixed request-time cash claim or a bridge-local queue fails the accepted case. The analogous **LBPVault** queue is a successor evidence profile and is not inferred from the current FIA-009 record.

6.10 FIA-010: exactly-once recovery

The canonical financial transaction confirms and a durable checkpoint is written. The worker terminates before projection completion. A fresh process replays the durable command, observes the completed receipt/state, performs no second financial transaction, and rebuilds the application and audit layers. The package retains both process identities and proves a financial transaction count of one.

6.11 FIA-011: mandatory capability subcases

FIA-011 is one parent assertion with two mandatory subcases.

FIA-011A: stale capability before admission. A quote or admission based on stale capability must reject before financial mutation. Retained evidence includes capability version and age, the exact causal freshness predicate, typed reason, before/after state, negative control, replay result, and a dedicated report row.

FIA-011B: book-epoch change mid-position while $D_t > 0$. The case begins with a position bound to book epoch b_0 and positive debt. A transition to $b_1 \neq b_0$ must revoke risk-increasing authority and produce the registered fail-closed or risk-reducing state. Evidence must bind the epoch transition causally to the terminal behavior and prove that no unregistered risk increase occurred.

Definition 6.1 (FIA-011 conjunctive closure). *The parent FIA-011 passes iff both FIA-011A and FIA-011B satisfy retained raw and normalized evidence, manifest-bound identity, exact transition provenance, causal checks, expected zero-mutation/rejection/fail-closed semantics, negative controls, replay, and explicit per-subcase report coverage.*

Schema presence, generated fixtures, one passing subcase, terminal-state equality, or an aggregate score cannot close FIA-011.

6.12 FIA-012: pause and recovery

A guardian command atomically pauses the manifest-frozen target set. Risk-increasing selectors fail; declared risk-reducing and recovery selectors remain available where safe. Governance queues and executes the timelocked recovery/unpause sequence. Partial pause, missing target, coordinator-local pause state, or direct unpause fails the case.

7 Additive venue and legal cases outside the FIA denominator

The top-level additions in table 2 remain outside the twelve-parent FIA set.

Pause policy. Under **PERSIST_THROUGH_PAUSE**, a resting order may support a certificate only when persistence remains risk-reducing. Under **CANCEL_ON_PAUSE**, an order is cancelled when persistence could increase risk and contributes zero certified capacity beyond the pause. Pause, reactivation reset, and post-close lock are distinct lifecycle states.

Table 2: Additive v0.2 cases and planning-label crosswalk.

Case ID	Planning label	Required behavior
VA-016	VA-TIMER-RESTART	settlement timer restart produces a new registered timing identity without rewriting the prior interval
VA-017	VA-BOOK-RESET	reactivation book reset increments epoch, invalidates stale capacity, and preserves the prior book as historical evidence
VA-018	VA-POST-CLOSE-LOCK	post-close mutation, cancellation, or risk increase is rejected with zero financial mutation
VA-019	VA-PAUSE-MODES	distinguishes trading pause from exchange pause and registers PERSIST_THROUGH_PAUSE and CANCEL_ON_PAUSE submodes
LG-001	LEGAL-NO-GO	displayed collateral or credit from an ineligible entity/custody/permitted-use class yields LEGAL_NO_GO and zero origination/risk increase

Non-retroactivity. If the current evidence run is already sealed, these cases are registered for a new additive run and do not alter prior run manifests or hashes.

8 Layer-specific schemas

The seven layers share a common envelope:

```
{
  "schema": "aemb/fia-layer/v0.2",
  "case_id": "FIA-001",
  "layer": "canonical_contract_state",
  "manifest_hash": "...",
  "correlation_id": "...",
  "source_inputs": [...],
  "derivation": {...},
  "projection": {...},
  "digest": "...",
  "observed_or_reconstructed": "OBSERVED",
  "tool_identity": {...}
}
```

The contract layer additionally requires terminal block, ABI views, ordered events, and target addresses. The journal layer requires append-only sequence and source event IDs. The API and browser layers require role/session class and route/control identity without retaining personal data. The audit layer requires a reconstruction input set distinct from the browser rendering artifact.

9 Reference verdict algorithm

The deterministic case procedure is stated in algorithm 1.

The algorithm returns one verdict and a complete gate vector. It never converts an absent layer to pass by default.

10 Case object and schema

Definition 10.1 (Abstract FIA case body). *An abstract FIA case body is the manifest-independent tuple*

$$C^0 = (v, i, u, x, E, N, Q),$$

Algorithm 1 AEMB v0.2 case evaluation**Require:** case C , manifest M , evidence package Z

```

1: verify package allowlist, member digests, schemas, and source identities
2: verify  $M$  and all manifest-bound fields
3: verify command, role, selector, receipt, and canonical state requirements
4: for each required layer  $L_i$  do
5:   verify retained inputs, derivation, digest, correlation, and semantic projection
6: end for
7: execute mandatory negative controls and require the registered rejection
8: verify clean-run, replay, and independent-review records when required by the profile
9: if any financial expectation is contradicted then
10:   return FAILED
11: else if a required authority/environment boundary prevents completion then
12:   return BOUNDARY_BLOCKED
13: else if only canonical replay exists then
14:   return CANONICAL_REPLAY_ONLY
15: else if read/reconstruction is complete but the financial command is absent then
16:   return READ_ONLY_E2E
17: else if all conjunctive gates pass then
18:   return FULL_E2E
19: else
20:   return BOUNDARY_BLOCKED with missing-gate registry
21: end if

```

where v is the schema version, i the case identifier, u the role, x the typed command, E the expected invariant object, N the negative-control registry, and Q the privacy/data-rights record. Deployment and source-release identities are excluded from C^0 .

Definition 10.2 (Case-manifest registration). A case-manifest registration binds the abstract body digest to a deployment-manifest digest and an evidence-policy version. Source-release provenance is stored in a separate registration object and never changes the abstract case-body identity.

A JSON representation is:

```

{
  "schema": "aemb/fia-case/v0.2",
  "case_id": "FIA-001",
  "role": "TRADER",
  "command": {...},
  "expected": {...},
  "actual_evidence_requirements": [...],
  "negative_controls": [...],
  "privacy_rights": {...}
}

```

A separate case-manifest object records:

```

{
  "case_body_digest": "...",
  "deployment_manifest_digest": "...",
  "evidence_policy_version": "aemb/evidence-policy/v0.2",
  "source_provenance_ref": "..."
}

```


The schema requires explicit nulls for inapplicable fields. Missing and inapplicable are not synonyms: a field is inapplicable only when the case definition says so; an absent required layer yields a blocked or failed verdict.

10.1 Source provenance registration

A separate provenance object identifies the benchmark version, case registry, reference implementation, contract release, emulator release, and source archive. It is not an input to the abstract case-body digest. This permits later releases to preserve one abstract case while evaluating multiple registered implementations.

10.2 Deployment manifest

The case-manifest layer references a manifest digest that binds the contract graph, code, roles, selectors, adapters, legal-authority classes, capability requirements, and application schemas. A pass under manifest M_1 is not automatically a pass under M_2 .

10.3 Expected object

The expected object contains:

- preconditions and rejection alternatives;
- expected financial invariants;
- permitted and forbidden mutation domains;
- expected terminal lifecycle class;
- expected simulator state or no-mutation condition;
- required evidence layers and negative controls.

The expected simulator hash is never an input to an actual evidence layer.

11 Identity hierarchy

The relationship among case, manifest, package, and release identities is summarized in fig. 1. AEMB v0.2 uses several identities because they answer different questions.

Definition 11.1 (Case-body identity). *Let $C^0 = (v, i, u, x, E, N, Q)$ and let JCS denote RFC 8785 canonical JSON. The manifest-independent case-body digest is*

$$h_C = H(\text{JCS}(C^0)).$$

Definition 11.2 (Manifest-bound case identity). *For manifest digest h_M and evidence-policy version p ,*

$$h_{CM} = H(\text{JCS}(h_C, h_M, p)).$$

Source-release provenance is registered separately and does not enter h_C or double-bind h_M .

Definition 11.3 (Evidence package identity). *For an ordered allowlisted set of evidence members $F = (f_1, \dots, f_k)$ with canonical paths and recorded metadata, define*

$$h_E = H(\text{archive}(F, \text{manifest}(F))).$$

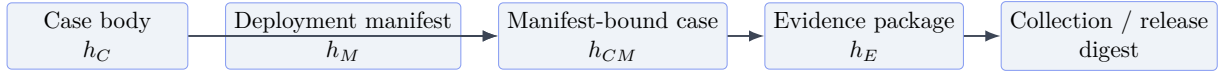


Figure 1: AEMB v0.2 identity hierarchy. The abstract case, tested deployment, retained evidence, and released dataset have different identities.

Definition 11.4 (Collection and release identities). *The FIA collection digest commits to the ordered list of manifest-independent case-body digests. The AEMB release digest commits to the v0.1 collections, the FIA collection, schemas, policy, metadata, and reference outputs.*

Separating case-body and manifest-bound identities allows the same abstract scenario to be evaluated against multiple approved implementations without pretending that their deployments are identical.

The release identity hierarchy is shown in fig. 1.

12 Seven retained evidence layers and derivation roots

Each FIA case declares seven retained layers:

1. simulator expectation;
2. canonical contract state;
3. append-only journal;
4. service projection;
5. authenticated API projection;
6. browser projection;
7. audit reconstruction.

A layer record contains:

$$L_i = (\tau_i, I_i, D_i, h_i, c, h_M),$$

where I_i is the retained input set, D_i the derivation, h_i the digest, c the correlation identity, and h_M the manifest digest.

Definition 12.1 (Seven-layer conformance). *A case is seven-layer conformant if every required layer is present, schema-valid, manifest-bound, correlated to the same command, derived from its retained inputs, and equal to the case’s canonical terminal semantics under the layer-specific projection.*

The benchmark does not require seven byte-identical objects and does not presume seven independent roots. It requires seven retained objects with registered inputs, derivations, root ancestry, and the same declared semantics. A canonical contract-state object and a browser object can differ structurally while yielding the same normalized protocol-state projection. Claims that assert independent runtime participation would additionally need a separately registered and evaluated multi-root corroboration predicate. The Phase 17B reference cohort did not evaluate that stronger predicate.

Proposition 12.2 (Equality without derivation is insufficient). *If all seven layer digests are copied from the simulator expectation, the case fails seven-layer conformance even though the digest values match.*

Definition 12.3 (Derivation-root record). *A derivation-root record identifies each retained root observation, every transformation edge, the runtime/implementation responsible for the edge, and the claims for which multi-root corroboration is required.*

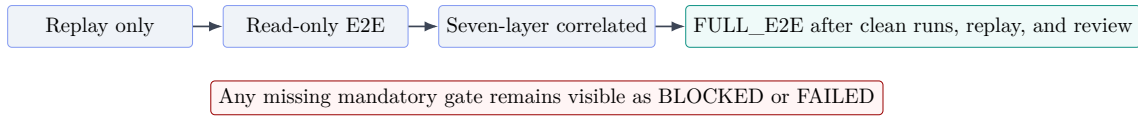


Figure 2: Verdict discipline. The path is not automatic; each arrow represents additional evidence and can fail closed.

Disguised common-root control. Proposed negative control NC-R00T-001 would repackage an L4 service object as an apparent L5 authenticated-API object. Its expected result is rejection because the root ancestry and derivation are unchanged; a new label does not create a new observation.

13 Predicate classes

Every gate in the registry is assigned exactly one class:

- MONOTONE_POSITIVE;
- TERMINAL_CLOSURE;
- ABSENCE_NEGATIVE_CONTROL;
- RECOMPUTED_COMPARISON;
- MULTI_ROOT_CORROBORATION.

Only MONOTONE_POSITIVE predicates are upward-closed under append-only valid evidence. Closure, absence, clean-run/replay comparison, and derivation-root predicates are terminal or recomputed. No theorem or validator may apply blanket monotonicity across all classes.

14 Verdicts and non-compensating evaluation

Per-case verdicts are:

- FULL_E2E: all required command, financial, projection, package, replay, and review gates pass;
- READ_ONLY_E2E: a complete read/reconstruction path exists without the required financial command;
- CANONICAL_REPLAY_ONLY: deterministic canonical replay exists without the full live local application path;
- BOUNDARY_BLOCKED: a declared authority, data-rights, environment, or policy boundary prevents completion;
- FAILED: required behavior or evidence is contradicted.

Definition 14.1 (Conjunctive case verdict). *Let $G(C)$ be the required gate set for case C . The case is FULL_E2E iff*

$$\bigwedge_{g \in G(C)} g = 1.$$

AEMB reports case and collection matrices, not one compensating safety score. An implementation may report performance, gas, or additional coverage separately, but those metrics cannot turn a failed evidence-conflict case into a pass.

15 Negative controls and mutation registry

AEMB v0.2 requires the mutation families in table 3.

Table 3: Mandatory negative controls.

Family	Examples
Manifest drift	wrong address, bytecode, ABI, storage layout, role, selector, adapter, schema, client
Receipt tamper	status, transaction digest, log/topic/data, chain ID, block/transaction/log order
State derivation	fixture hash copied into actual layer; harness storage treated as canonical; omitted ABI view
Journal/projection	missing, duplicate, reordered, stale, or copied record
Correlation	wrong role/session, command, idempotency domain, evidence ID, transaction, or scenario
Settlement	duplicate evidence, conflicting evidence, amount beyond matched capacity
Waterfall/queue	wrong priority, non-conservation, queue service before loss participation, double-counted capital
Recovery	second financial transaction, missing checkpoint, same process presented as restart
Governance	partial pause, target omission, direct unpause without timelock, recovery path disabled
Package	path traversal, member injection, missing file, checksum self-reference, nondeterministic executable, secret/PII
Boundary	external write, remote RPC, production asset/config, public liquidity, leverage above declared scope

Mutation adequacy is reported case-wise. A surviving mutation is not hidden by aggregate detection rates; it becomes coverage debt for the next benchmark revision [Jia and Harman, 2011]. Differential canonicalizers and validators should be implemented in multiple languages [McKeeman, 1998].

16 Prospective v0.1-to-v0.2 coverage map

AEMB v0.1.0 is not relabeled retroactively. The following six uncovered financial-safety families map prospectively to registered v0.2 negative-control families.

The mapping states what v0.2 is designed to add. It does not change the registered scope or results of AEMB v0.1.0.

Table 4: Prospective coverage of the six v0.1 uncovered financial-safety families.

Control ID	Previously uncovered family	v0.2 expected rejection
NC-FIN-001	settlement above matched capacity	cap violation rejected; debt and residual unchanged
NC-FIN-002	interest or fee before debt under insufficient proceeds	debt-first priority enforced; unauthorized senior payment rejected
NC-FIN-003	wrong reserve/LBP/Senior ordering or protection-capital reuse	ordering/conservation/reuse gate fails
NC-FIN-004	zero-collateral origination	origination rejected with zero position and debt
NC-FIN-005	risk increase while debt remains positive	risk-increasing command rejected with zero mutation
NC-FIN-006	attempted external venue write	environment boundary rejects the command and records zero external writes

17 Reference implementation architecture

The reference release should include:

- JSON Schemas for cases, layers, packages, verdicts, and release metadata;
- canonical JSON implementation and test vectors;
- Python and Node.js validators plus at least one independent implementation;
- manifest verifier;
- canonical contract-state collector;
- journal, service, API, browser, and audit normalizers;
- mutation generators;
- package builder and allowlist verifier;
- historical replay wrapper;
- Dataset Card, CITATION metadata, SBOM, license files, and correction policy.

The reference code is not the benchmark definition. A correct independent implementation may use different internal structure, provided it reproduces the declared case semantics and evidence predicates.

18 Clean runs, replay, and independent review

18.1 Two clean packages

A final candidate is built twice in independently created worktrees and runtime directories. The comparison record distinguishes:

- raw fields that must be byte-identical;
- runtime fields that may differ but are retained;
- normalized semantic fields that must match;
- fields excluded from normalization with explicit justification.

A statement of byte identity is allowed only when the archive bytes actually match.

18.2 Historical replay

The release records the provenance chain from earlier phases to the FIA evidence. If an early phase lacked a frozen replay wrapper or expected output hash, a later replay is labeled a post-hoc reproducibility attestation rather than a retroactive historical identity.

18.3 Independent review

A separate path verifies:

- (1) package manifest and member allowlist;
- (2) schemas and canonicalization;

- (3) case and collection digests;
- (4) manifest binding;
- (5) layer inputs and derivations;
- (6) negative controls;
- (7) clean-run comparison;
- (8) replay provenance;
- (9) privacy, rights, and secret boundaries;
- (10) final verdict matrix.

The reviewer may disagree with a policy choice while still verifying that the evidence package implements the declared policy.

19 Relationship to AVET

AVET v0.2.0 contains synthetic, source-generated venue-emulator traces [Nechepurenko, 2026b]. AVET and AEMB serve different roles:

- AVET preserves ordered venue-side histories, provenance, clocks, faults, and recovery traces;
- AEMB defines compact conformance cases with expected invariants and evidence requirements.

An AVET trace may motivate or parameterize an AEMB FIA case, but raw third-party venue data is not required. The public release should use synthetic or rights-cleared material and preserve source-specific rights metadata.

20 Reference Phase 17B corpus and observed result

Reference corpus available. The Phase 17B evidence companion now supplies one twelve-parent FIA cohort, 84 retained evidence positions, two matching clean materializations, a strict archive accepted by a separate verifier, and a selected P01–P17 replay bound to the archive. The standalone article is evidence-integrated; the public AEMB v0.2 dataset/schema release remains an owner-controlled freeze.

The public reference metadata are available in the publication companion [AxientLab, 2026]. The completed reference profile is summarized in table 5. The reference cohort is identified by run `phase17b-run-4c40d373b9c4ead91ecfdcab5539941743733b06`, source commit `bd6b2d54ec6e897b49756399637aaa49647b8ce1`, manifest SHA-256 `342b2a6f27aeb91bba2013f5604ab1c50aaba48d6f218c3f3f84cc5c39ce7bfd`, and ledger root `09cd709444eb82a0c2c17d4b42a2750b2cd9b6002f69b6a3648c4e2ea1b41e46`. The cohort contains twelve parent FIA records exactly once and seven retained `ACTUAL_EVIDENCE` positions per record.

The individual scenario-record identities and correlations are registered in the reference all-12 manifest. The public verifier recomputes repository-file checksums and validates schemas, counts, ordering, and internal consistency; it does not reconstruct the deliberately unpublished raw runtime payloads or independently recompute their payload-derived canonical identities. The clean materialization profile uses separate source reads; the package profile uses a source-owned producer and separate verifier; and the replay profile preserves selected predecessor/version edges before binding P17 to the archive.

Table 5: Reference FIA profile result.

Measure	Observed result
Parent FIA cases	12; FIA-011A/B mandatory inside FIA-011
Evidence positions	84
Clean materializations	two; one 16-file, 122,636-byte file index
Comparison SHA-256	53ebd95f...b3447187
Strict archive	16 members, 88,569 bytes; SHA-256 6901e327...e54cf328
Independent archive review	accepted by separate fixed-metadata verifier
P01–P17 replay	17 callbacks; result 2e51296a...18bb0767f
P17 archive binding	output 111a9c6c...fa6cda1b; binding 7363650e...3babba64
Source-owned adapter tests	source workspace reported 15 passed, 0 failed; publication companion does not recompute the suite

20.1 AEMB v0.2 release status

These observed identities populate the standalone article and reference-profile design. The public dataset release still requires a frozen additive registry, case/schema files, deterministic public package builds, independent review, licenses, citation metadata, and an owner-approved immutable tag. Existing AEMB v0.1.0 identities remain unchanged.

21 Release package layout

A recommended public/private package layout is:

```
AEMB-v0.2/
dataset/
  cases/FIA-001.json ... FIA-012.json
  cases/VA-016.json ... VA-019.json
  cases/LG-001.json
  subcases/FIA-011A.json FIA-011B.json
  collections/fia.json
  controls/NC-ROOT-001.json NC-FIN-001.json ... NC-FIN-006.json
  schemas/
reference/
  python/
  node/
  independent-verifier/
evidence-policy/
  layer-schema.json
  verdict-schema.json
  package-schema.json
  mutation-registry.json
controls/
  malformed/
  mutations/
metadata/
  DATASET_CARD.md
  CITATION.cff
  codemeta.json
  zenodo.json
  RIGHTS_AND_PRIVACY.md
```



```
reports/  
  CLEAN_RUN_COMPARISON.json  
  REPLAY_ATTESTATION.json  
  INDEPENDENT_REVIEW.md  
MANIFEST.sha256  
SHA256SUMS.txt
```

The public dataset need not include confidential contract source or raw operational logs. It must include enough retained, sanitized evidence to verify the published conformance result.

22 Coverage accounting

AEMB v0.2 reports the dimensions summarized in table 4:

- FIA parent coverage: 12 registered cases, with FIA-011A/B mandatory inside FIA-011;
- additive top-level coverage: four venue-capability cases and one legal no-go case;
- negative-control coverage: derivation-root control plus six prospective financial-safety families;
- authority coverage: risk, position, debt, evidence, capital, liquidation, governance, venue capability, legal entity, custody, and permitted use;
- evidence-layer coverage: seven layers per case where applicable;
- negative-control coverage: mutation families and individual controls;
- language/runtime coverage: reference and independent validators;
- replay coverage: historical phases and current package;
- review coverage: mechanical and semantic checks.

The required-versus-observed coverage dimensions are summarized in table 4. Coverage is not completion. A layer may be represented in the schema but missing from the actual package; that is reported as coverage debt. The release should include a matrix with required, present, passed, and independently reviewed counts.

23 Privacy, security, and data-rights boundary

The FIA collection is synthetic/local and should contain no participant identity, wallet address, credential, private key, remote RPC, production configuration, raw third-party venue payload, or live financial data. Browser evidence may use an anonymized role/session class rather than a user identifier. Operational logs are excluded unless converted into a minimal allowlisted evidence record.

A secret/PII scan is necessary but not sufficient: reviewers must also inspect whether an innocuous field can expose internal topology, private repository paths, or sensitive incident details. Public evidence diagrams and tables should be derived from the sanitized package rather than copied from an active developer environment.

24 Publication and disclosure boundary

This article reports the public Phase 17B reference corpus and the proposed additive benchmark specification. The pinned companion commit is `7722146bee0e7f02d0c4f9f63ada6d527e6ff079`. The AEMB v0.2 dataset, schemas, validators, and package identities form a separate versioned release.

That release should expose only sanitized synthetic cases, declared evidence requirements, deterministic validators, and the applicable license/trademark boundaries; raw operational payloads and unrelated implementation details are outside the dataset scope.

A Normative per-case checklist

For each FIA case the release checklist includes:

- (1) immutable case body and digest;
- (2) source release and manifest identity;
- (3) role and command schema;
- (4) preconditions and expected financial invariants;
- (5) permitted and forbidden mutation domains;
- (6) seven-layer requirement vector;
- (7) negative-control registry;
- (8) reference and independent validator outputs;
- (9) clean-run and replay references;
- (10) privacy/rights classification;
- (11) final verdict and missing-gate list;
- (12) correction/deprecation metadata.

B Reference case status

The public companion records each FIA parent case in the accepted cohort; table 6 reports their reference-profile inclusion. This table reports reference-profile inclusion; it is not a universal safety score.

Table 6: Reference FIA status.

Case	Reference predicate	Observed profile status
FIA-001	admission/origination	accepted cohort record; seven retained positions
FIA-002	exact-2× admission	accepted cohort record; distinct correlation
FIA-003	over-cap rejection	accepted no-mutation record
FIA-004	partial/final settlement	accepted settlement record
FIA-005	residual/collateral release	accepted release record
FIA-006	liquidation to hard-flat	accepted liquidation record
FIA-007	reserve-LBP-Senior waterfall	accepted allocation record
FIA-008	conflict/quarantine/recovery	accepted exception/recovery record
FIA-009	withdrawal queue under loss	accepted queue record
FIA-010	exactly-once replay/recovery	accepted replay record
FIA-011	mandatory A/B capability predicate	accepted two-subcase parent record

Case	Reference predicate	Observed profile status
FIA-012	pause/timelocked recovery	accepted governance record

C Versioning and governance

Version 0.2 is a MINOR additive release if v0.1.0 case semantics and digests remain unchanged. A change to v0.1 canonicalization, expected terminal state, or identity semantics requires a separate correction decision. Published files are immutable; substantive changes create a new version rather than silently replacing the archive.

Licensing should preserve the existing split unless the owner decides otherwise:

- scenario data and public outputs: CC BY 4.0;
- schemas, canonicalizers, validators, and reference code: Apache-2.0;
- Axient names and marks: excluded from open grants;
- raw or derived third-party venue data: no automatic redistribution.

The exact protocol implementation license must be decided separately from dataset licensing.

D Relation to benchmark and testing literature

AEMB combines property-based testing [Claessen and Hughes, 2000], differential testing [McKeeman, 1998], mutation analysis [Jia and Harman, 2011], smart-contract fuzzing [Wüstholtz and Christakis, 2020], canonical serialization [Rundgren et al., 2020], and supply-chain provenance [Torres-Arias et al., 2019, Open Source Security Foundation, 2023]. Its distinct contribution is a financial-protocol conformance object in which a case includes not only an expected state but also a manifest-bound, cross-layer evidence obligation and a non-compensating verdict.

E Limitations

AEMB v0.2 cannot certify production safety, economic profitability, legal compliance, oracle truth, venue solvency, key security, or the absence of unknown vulnerabilities. Passing a local FIA case proves only the declared local, emulated scope. Reference validators can share common bugs. Independent review reduces but does not eliminate that risk. The benchmark also reflects the selected financial architecture and evidence policy; other protocols may require additional cases or different authority maps.

F Conclusion

AEMB v0.2 extends deterministic protocol testing from terminal-state comparison to manifest-bound end-to-end conformance. Its twelve parent FIA cases, mandatory FIA-011A/B subcases, and additive venue/legal cases make admission, settlement, liquidation, queues, recovery, capability transitions, legal eligibility, and governance reviewable across seven retained evidence layers with registered derivations. The benchmark’s central rule is conservative: no missing safety-critical layer can be hidden by extra passing tests or matching copied hashes. The reference profile is now backed by the completed Phase 17B evidence chain. The additive public AEMB v0.2 dataset remains a separate versioned release requiring schema/package freeze and independent review.

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